

Cem Özçelik

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PROFESSIONAL SUMMARY

Senior Data Scientist and AI Engineer with nearly 5 years of end-to-end machine learning experience and an Industrial Engineering foundation. I build production Agentic AI systems with LangChain and LangGraph, where LLMs route to specialized model pipelines and Operations Research solvers return optimal, auditable decisions rather than plausible ones.

Shipped a Multi-Agent demand-forecasting framework and a LangGraph ERP conversational copilot to production, plus semantic retrieval over 800K+ documents on a Qdrant vector database (the retrieval foundation of RAG). Strong across the full ML lifecycle and MLOps: feature engineering, model training and tuning, monitoring, and deployment with Docker, FastAPI, and Django REST APIs.

Insurance is a natural fit for my cost-function mindset: every decision is a cost-asymmetry problem (false claim approval vs missed fraud), the same shape I solved in Credit Risk (XGBoost, class-imbalanced) and clinical diagnostics (97% Recall).

CORE COMPETENCIES

End-to-End ML Lifecycle

Multi-Agent Systems (LangChain/LangGraph)

RAG & Vector Databases (Qdrant/Pinecone)

MLOps & Model Monitoring

LLM Orchestration & Prompt Strategies

Statistical Modelling & Data Mining

Insurance / Risk Modelling

Python & SQL at Scale

WORK EXPERIENCE

Anadolu Pet Akvaryum & Effeffe

Jan 2026 - Present

Senior Database Analyst - Maltepe, Istanbul

- Agentic AI & Demand Forecasting:** Led a cross-functional team of four to architect a **Multi-Agent decision-support framework (LangChain)** automating rolling ABC-XYZ demand segmentation. Engineered a **Forecaster Agent** routing SKUs to specialized pipelines (JIT for AX, XGBoost for BY, ML-driven safety-stock optimization for CZ), sustaining an 8-hour weekly reduction in operational overtime and improving inventory turnover.
- LLM + Operations Research fusion:** Implemented a **LangGraph-based route profitability engine** that integrates OR mathematical solvers inside the multi-agent workflow to compute real-time "True Cost to Serve," converting high-overhead orders into profitable routes (targeting +10% net logistics margin).
- MLOps & ML infrastructure:** Scaled a custom Django application into a centralized hub for **deploying ML models via RESTful APIs**; translated SKU-level Association Rule Mining into business actions driving continuous revenue growth.

Anadolu Pet Akvaryum & Effeffe

Sept 2024 - Dec 2025

IT & Data Analytics Specialist - Maltepe, Istanbul

- Conversational AI / RAG copilot:** Designed an interactive **AI assistant using LangChain and LangGraph** to streamline a manual quarterly ERP budget workflow. Built a multi-node state-machine framework integrated with **Oracle DB** to validate live user entries against ERP constraints at each step, mitigating transactional errors for non-technical business units.
- Predictive supply chain & analytics:** Developed predictive supply-chain optimization models in Python and PL/SQL (reducing stockouts, improving turnover); ran route-level profitability analysis in advanced SQL for a 3% logistics-efficiency gain.
- Full-stack engineering:** Architected and deployed a centralized internal web application from scratch with Python (Django) and SOLID principles, establishing the company's foundational software engineering practices.

Birinci Automotive

Jan 2022 - Sept 2024

Mid-Level Data Analyst - Çayırova, Kocaeli

- **Predictive Maintenance & sensor analytics:** Built a Predictive Maintenance framework targeting the OEE Availability metric using unsupervised clustering on vibration, temperature, and electrical sensor time-series, delivering a 15% weekly increase in critical machine availability and a foundation for sequential (RNN) modelling.
- **Computer Vision for quality control:** Deployed a **YOLO computer-vision pipeline** monitoring the OEE Quality metric on hot-metal forge-press conveyors; overcame high-glare conditions with advanced Data Augmentation, improving defect-detection accuracy.
- **Data engineering & OEE optimization:** Engineered robust ETL pipelines in MSSQL and Python, replacing manual reporting and driving an overall 15% improvement in Overall Equipment Effectiveness.

Birinci Automotive

June 2021 - Jan 2022

Junior Data Analyst - Çayırova, Kocaeli

- Adapted the CRM RFM framework to manufacturing equipment, segmenting machinery by maintenance recency, breakdown frequency, and repair cost to optimize maintenance scheduling and save 8 hours (one full shift) of downtime.
- Conducted EDA and statistical analysis on workforce datasets, aligning productivity metrics with manufacturing OEE goals; built automated ingestion/ETL workflows that improved reporting efficiency by 10%.

PROJECTS

E-Commerce AI Search Engine - Semantic Search, Vector DB & RAG Retrieval

Production microservices with dual-engine A/B testing: SentenceTransformers (all-MiniLM-L6-v2) vs custom Word2Vec over 800K+ product descriptions, backed by a **Qdrant vector database** (same API surface as Pinecone/Milvus). Item2Vec collaborative filtering for recommendations; custom LSTM next-token model for search auto-complete; sub-millisecond cached responses via Redis; PySpark distributed ETL. The retrieval architecture maps directly to insurance RAG (quote comparison, claims Q&A, policy chatbots).

PyTorch · FastAPI · Qdrant · Redis · PySpark · SentenceTransformers · Docker · github.com/ccemozclk/amazon-ai-search-engine

End-to-End Credit Risk Prediction - Risk Modelling & MLOps

Full MLOps lifecycle on the Lending Club dataset optimized for heavy class imbalance with temporal holdout validation (no leakage): raw ingestion → feature engineering → model training → containerized production app. Interactive Django EDA platform for non-technical reviewers. The probability-of-event-under-asymmetric-cost pattern transfers 1:1 to insurance claim, fraud, and churn modelling.

Python · XGBoost · CatBoost · Django · Docker · Scikit-learn · github.com/ccemozclk/Credit-Risk

Retail Demand Forecasting - End-to-End MLOps Pipeline

~12.83% RMSPE on 6-week daily sales forecasting across 1,115 stores with time-based validation. Object-oriented training pipeline (Ingestion → Transformation → ModelTrainer), FastAPI inference, and a Streamlit dashboard translating ML metrics into business terms (€ margin of error). Resilient inference layer handling unseen categorical inputs in production.

Python · LightGBM · XGBoost · Optuna · FastAPI · Streamlit · Docker · github.com/ccemozclk/Rossmann-Demand-Forecasting

EDUCATION

Kırıkkale University - B.Sc. Industrial Engineering

Sept 2016 - Feb 2021

Mathematical foundation in statistics, linear algebra, and optimization. Built MILP models with IBM ILOG CPLEX and applied Multi-Criteria Decision Making (AHP, TOPSIS) - direct roots of cost-function thinking.

CERTIFICATIONS

Associate Data Scientist Career Track (PySpark, Feature Engineering, ALS)	Datacamp	2024
Data Science & Machine Learning Bootcamp (6-month, end-to-end ML)	Miuiul	2022

SKILLS

Agentic AI & LLMs: LangChain, LangGraph, Multi-Agent Systems, RAG, Prompt Engineering, State-Machine Orchestration, Vector Databases (Qdrant)

AI & Machine Learning: Scikit-learn, XGBoost, CatBoost, h2o.ai, Time-Series Forecasting, Anomaly Detection, Association Rule Mining, Clustering (K-Means, DBSCAN)

Deep Learning & CV: TensorFlow, Keras, PyTorch, YOLO, CNNs, RNNs/LSTMs, Grad-CAM (Explainable AI), Transfer Learning

MLOps & Deployment: Docker, Django, FastAPI, Flask, RESTful APIs, Git, Linux, Nginx, SOLID

Data & Big Data: Python (Advanced), SQL (PL/SQL, MSSQL, Oracle, PostgreSQL), Apache Spark (PySpark), Databricks, ETL, Pandas, NumPy

IE & Optimization: MILP, AHP/TOPSIS, OEE Optimization, Supply Chain (ABC-XYZ), Inventory Theory (JIT, Safety Stock)

Languages: Turkish (Native), English (Fluent), Spanish (Basic)